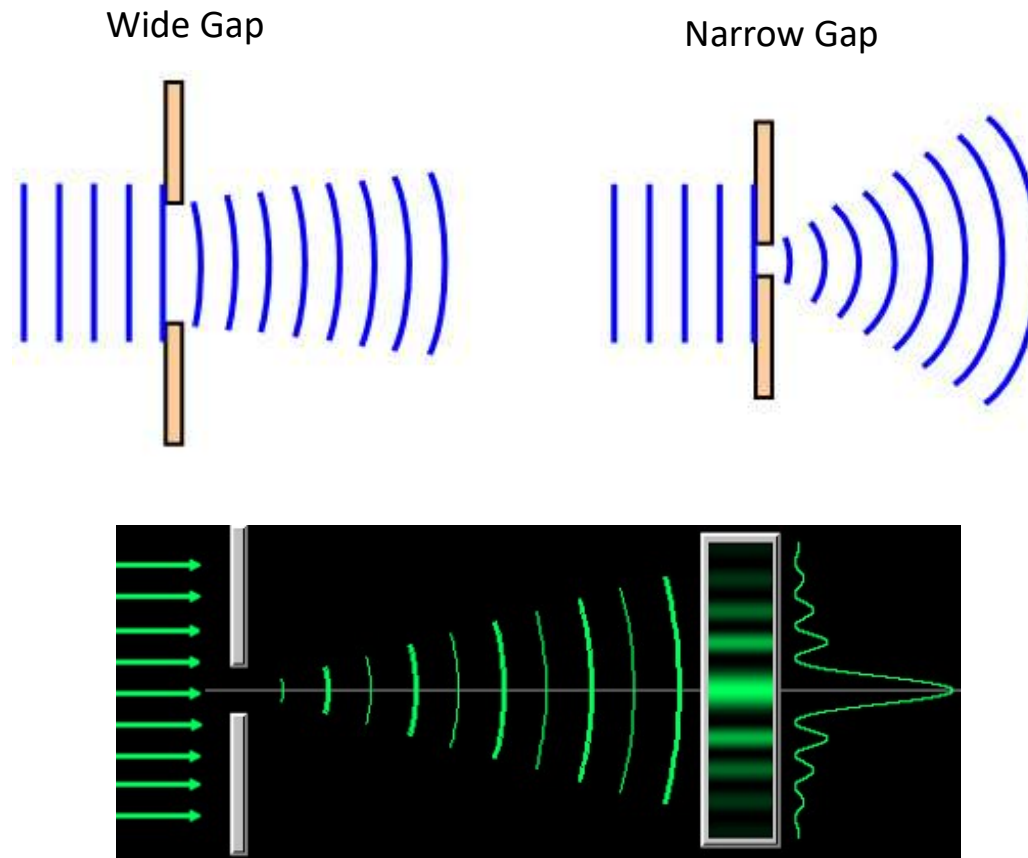


Diffraction of Light

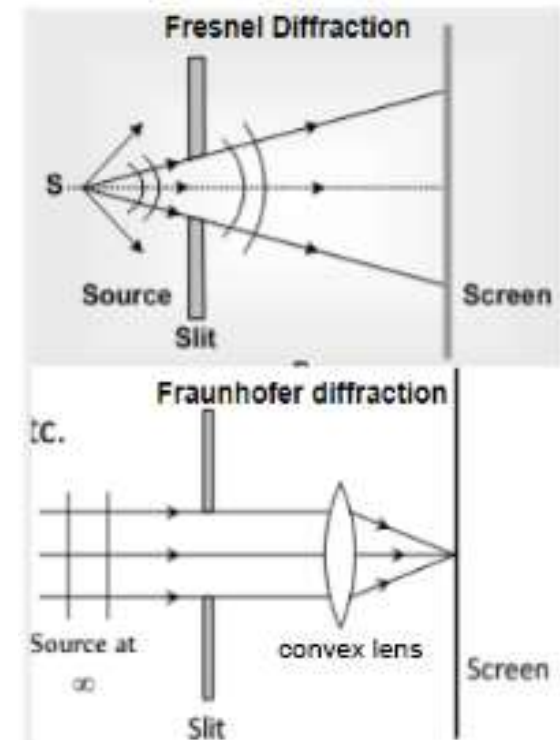
The spreading or bending of a light wave when it passes through a narrow opening is known as diffraction of light. The intensity distribution on screen is known as diffraction pattern.



Types of diffraction:

- Fresnel diffraction (near field diffraction)
- Fraunhofer diffraction (far field diffraction)

Fraunhofer diffraction	Fresnel diffraction
Source and the screen are far away from each other.	Source and screen are not far away from each other.
Incident wave fronts on the diffracting obstacle are plane.	Incident wave fronts are spherical.
Diffracting obstacle give rise to wave fronts which are also plane.	Wave fronts leaving the obstacles are also spherical.
Plane diffracting wave fronts are converged by means of a convex lens to produce diffraction pattern.	No Convex lens is needed to converge the spherical wave fronts.



Fresnel number, F

Fraunhofer diffraction occurs when:

$$F = \frac{h^2}{d\lambda} \ll 1$$

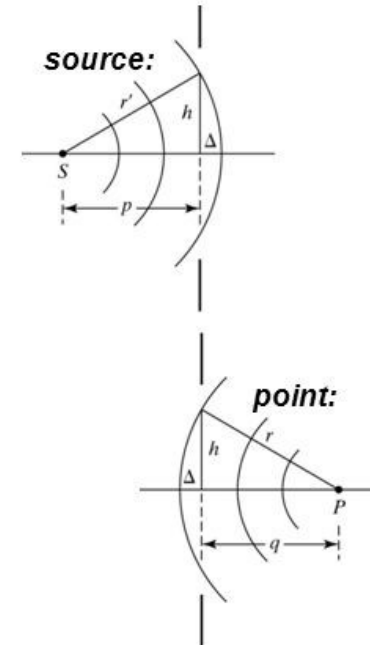
Fresnel diffraction occurs when:

$$F = \frac{h^2}{d\lambda} \geq 1$$

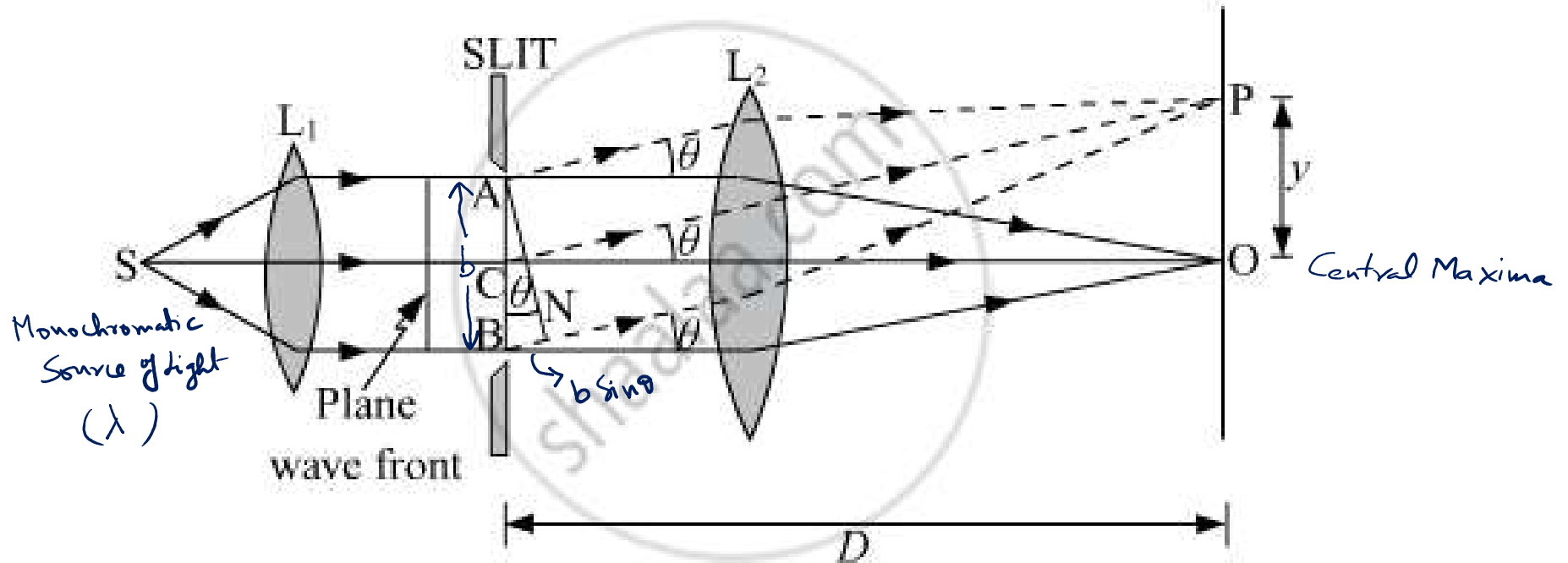
where h = aperture or slit size

λ = wavelength

d = distance from the aperture (p or q)



Single slit diffraction



Path difference between the waves emerging out at extreme points A & B of the slit = $BN = b \cdot \sin \theta$

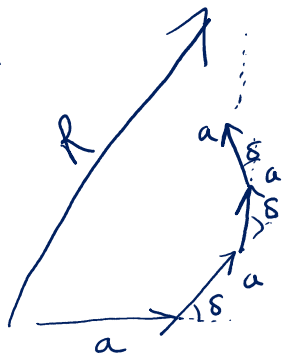
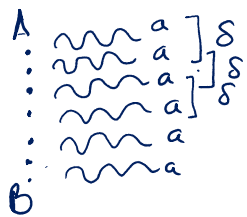
$$\text{Phase difference} = \frac{2\pi}{\lambda} \cdot \text{Path difference}$$

$$= \frac{2\pi}{\lambda} \cdot b \sin \theta$$

Suppose slit AB is divided into n equal parts
 Phase diff. between two consecutive parts

$$\delta = \frac{1}{n} \cdot \frac{2\pi}{\lambda} \cdot b \sin \theta$$

Suppose amplitude of each wavelet is 'a'.



Resultant Amplitude of all such waves (Wavelets) reaching at point P on the screen,

$$R = \frac{a \cdot \frac{\sin \frac{n\delta}{2}}{\frac{\delta}{2}}}{\sin \frac{\delta}{2}}$$

$$= a \cdot \frac{\sin \left(\cancel{n} \cdot \frac{1}{\cancel{n}} \times \frac{2\pi}{\lambda} \times \frac{b \sin \theta}{2} \right)}{\sin \left(\frac{\pi b \sin \theta}{n\lambda} \right)}$$

$$= a \frac{\sin \left(\frac{\pi b \sin \theta}{\lambda} \right)}{\sin \left(\frac{\pi b \sin \theta}{n\lambda} \right)}$$

$$\text{Let } \frac{\pi b \sin \theta}{\lambda} = \alpha$$

$$R = a \cdot \frac{\sin \alpha}{\sin \left(\frac{\alpha}{n} \right)}$$

As n is large, $\frac{\alpha}{n} \rightarrow \text{small}$, $\sin \left(\frac{\alpha}{n} \right) \approx \frac{\alpha}{n}$

$$R = a \cdot \frac{\sin \alpha}{\alpha/n}$$

$$R = ma \cdot \frac{\sin \alpha}{\alpha}$$

$$\text{if } na = A$$

$$R = A \cdot \frac{\sin \alpha}{\alpha}$$

Resultant intensity

$$I = R^2 = A^2 \cdot \frac{\sin^2 \alpha}{\alpha^2}$$

I is the intensity of ^{diffracted} light reaching at point 'P' on the screen.

Now we have to see, in which condition intensity I will be maximum or a minimum.

Maximum and minimum intensity,

$$\frac{dI}{d\alpha} = 0$$

$$\frac{d}{d\alpha} \left(A^2 \cdot \frac{\sin^2 \alpha}{\alpha^2} \right) = A^2 \cdot \frac{2 \sin \alpha}{\alpha} \cdot \frac{\alpha \cos \alpha - \sin \alpha}{\alpha^2}$$

It gives either,

$$\frac{\sin \alpha}{\alpha} = 0 \quad \text{or} \quad \frac{\alpha \cos \alpha - \sin \alpha}{\alpha^2} = 0$$

$$\frac{\sin \alpha}{\alpha} = 0 \quad \text{or} \quad \alpha = \tan \alpha$$

Direction of minimum intensity,

$$\frac{\sin \alpha}{\alpha} = 0 \quad \text{or} \quad \sin \alpha = 0$$

$$\alpha = \pm m\pi \quad \text{when } m=1, 2, 3, \dots$$

$$\frac{\pi b \sin \theta}{\lambda} = \pm m\pi$$

$$\therefore \boxed{b \sin \theta = \pm m\lambda}, \quad m=1, 2, 3, \dots$$

When, $m=0$, $b \sin \theta = 0$, this is condition for maximum intensity.

above equation gives direction of first, second, third... minima by putting $m=1, 2, 3, \dots$ respectively.

Directions of minimum intensity (Single slit)

$$\alpha = \tan \alpha$$

Graph plotting $y = \alpha$
 $y = \tan \alpha$

The points of intersection of the two curves give the required values of α .

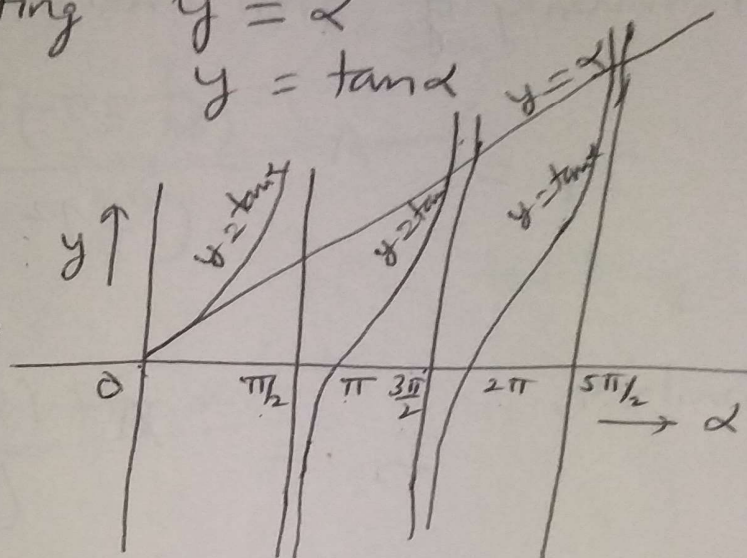
First value of $\alpha = 0$

Other values

$$\frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \dots \text{ (Approx.)}$$

Exact values

$$\alpha = 0, 1.43\pi, 2.46\pi, 3.47\pi, 4.47\pi \dots$$



The value $\alpha = 0$ corresponds to position of central maximum.

$$\frac{\pi b \sin \theta}{\lambda} = 0 \quad \text{or} \quad \theta = 0$$

Intensity $I = R^2 = A^2 \frac{\sin^2 \alpha}{\alpha^2} = A^2$
 $= I_0$ (say)

The direction of secondary maxima

$$\alpha = \pm \frac{(2m+1)\pi}{2}, \quad m = 1, 2, 3, \dots$$

or, $b \sin \theta = \pm (2m+1) \frac{\lambda}{2}$

For $m = 1, 2, 3, \dots$ we get position of
second, third ... secondary maxima when

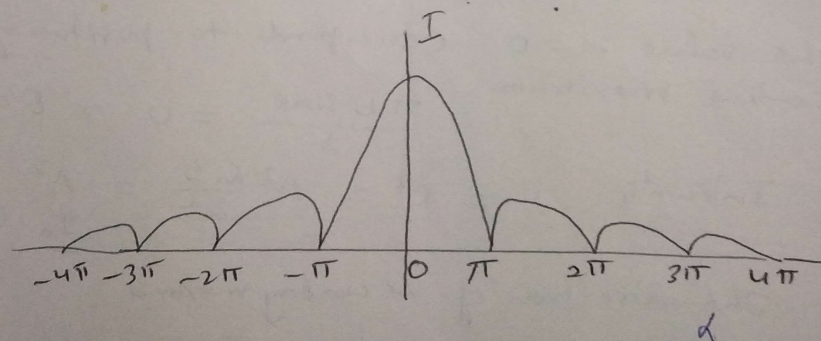
Intensity of first secondary maxima

$$I_1 = A^2 \frac{(\sin \frac{3\pi}{2})^2}{(\frac{3\pi}{2})^2} = \frac{A^2}{2^2} = \frac{I_0}{2^2}$$

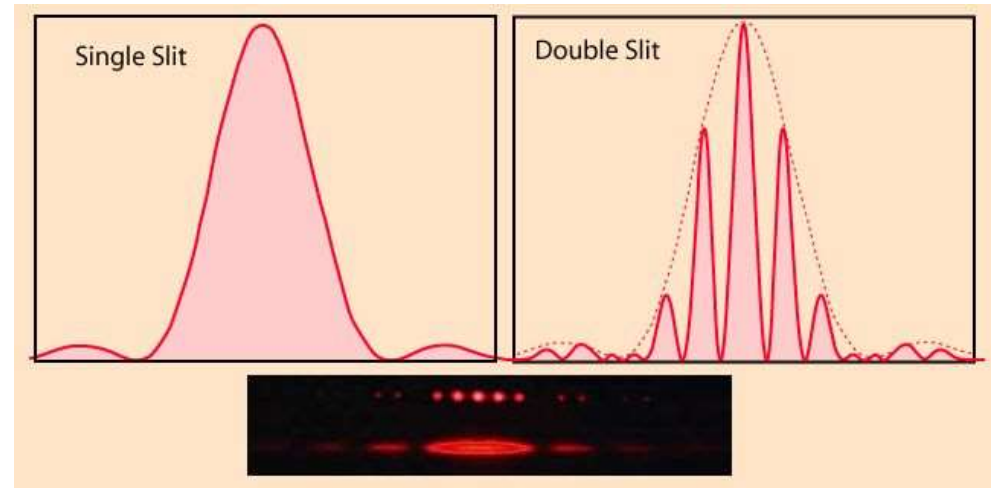
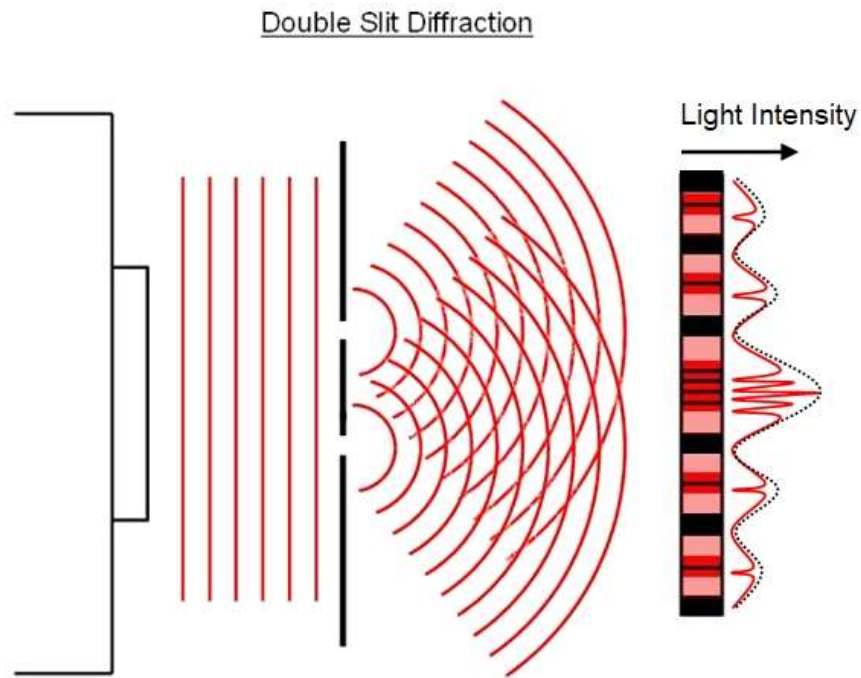
Similarly,

$$I_2 = A^2 \frac{(\sin \frac{5\pi}{2})^2}{(\frac{5\pi}{2})^2} = \frac{A^2}{5^2} = \frac{I_0}{5^2}$$

$$I_3 = \frac{I_0}{12^2}$$

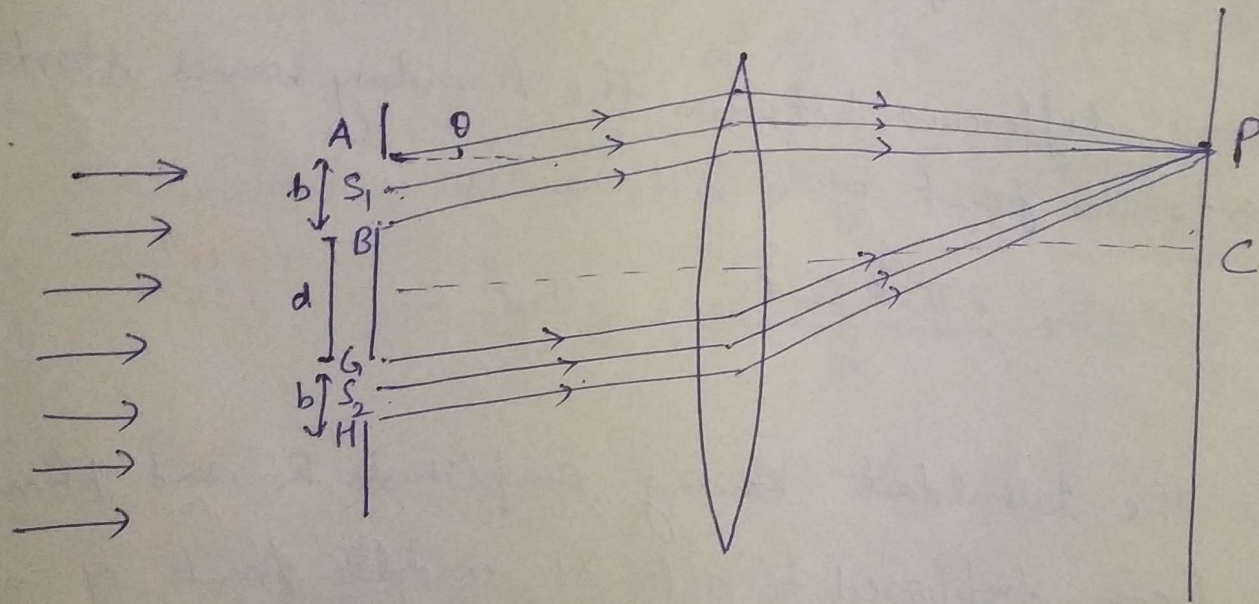


Two slit diffraction



Two slit diffraction pattern is formed due to,

- (i) Single slit diffraction due to individual slit
- (ii) Interference between two slits

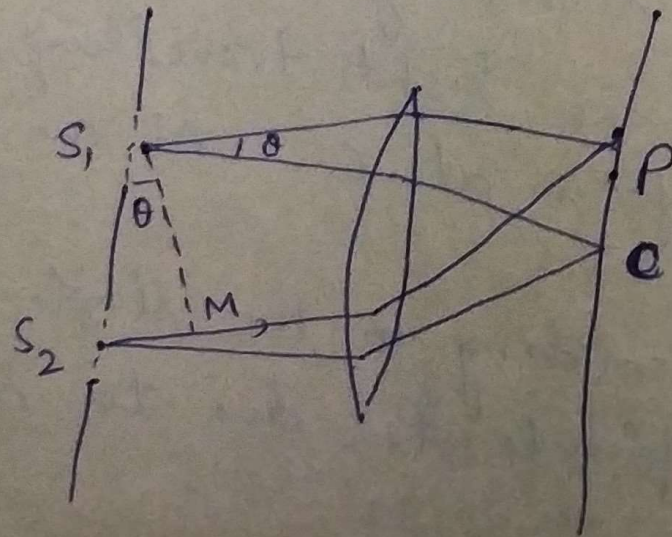


When a plane wave front is incident normally on the slits, all points within the slit become the sources of secondary wavelets which travel in all directions. All the light travelling along incidence direction come to focus at C. At an angle θ , the light focus at P.

Therefore the resultant having amplitude R and phase α has been supposed to act at middle points of slits. The resultant amplitude at P will be due to interference between the two waves of same amplitude R and a phase diff. ϕ .

$S_1 M$ is $\perp$ to $S_2 M$

The path diff. between waves reaching at P from S_1 & S_2
 $= S_2 M$

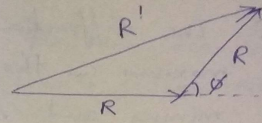


$$S_2 M = (b+d) \sin \theta$$

The phase difference,

$$\phi = \frac{2\pi}{\lambda} \cdot (b+d) \sin \theta$$

From triangle, the resultant amplitude, at P is given by,



$$\begin{aligned} R'^2 &= R^2 + R^2 + 2R \cdot R \cos \phi \\ &= 2R^2 (1 + \cos \phi) \\ &= 2R^2 \cdot 2 \cos^2 \phi/2 = 4R^2 \cos^2 \phi/2 \end{aligned}$$

Substituting value of R and ϕ , we get:

$$R'^2 = 4 \times A^2 \cdot \frac{\sin^2 \alpha}{\alpha^2} \cdot \cos^2 \phi/2$$

$$\text{Let } \phi/2 = \beta$$

$$R'^2 = 4 A^2 \frac{\sin^2 \alpha}{\alpha^2} \cdot \cos^2 \beta$$

Intensity at P,

$$I = 4 \cdot A^2 \cdot \frac{\sin^2 \alpha}{\alpha^2} \cdot \cos^2 \beta$$

- (i) factor $A^2 \frac{\sin^2 \alpha}{\alpha^2}$ shows a single slit diffraction pattern.
- (ii) the factor $\cos^2 \beta$ gives interference pattern.

The resultant intensity at any point on the axis is given by product of these two factors and will be zero when either of these factors is zero.

- (a) The diffraction term, $A^2 \frac{\sin^2 \alpha}{\alpha^2}$ gives central maxima in the direction $\theta = 0$, having alternate minima and secondary maxima of decreasing intensity on either side.

The angular position of minima are given by,

$$\sin \alpha = 0, \alpha \neq 0 \quad \lim_{\alpha \rightarrow 0} \frac{\sin \alpha}{\alpha} = 1$$

$$\alpha = \pm m\pi, m = 1, 2, 3, \dots$$

$$\frac{\pi b \sin \theta}{\lambda} = \pm m\pi$$

$$\sin \theta = \pm \frac{m\lambda}{b}, m = 1, 2, 3, \dots$$

The position of secondary maxima,

$$\alpha = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots \text{ (approx.)}$$

- (b) According to second factor $\cos^2 \beta$, the intensity will be maximum, when

$$\cos^2 \beta = 1$$

$$\beta = \pm n\pi \text{ where, } n = 0, 1, 2, 3, \dots$$

$$a, \quad \frac{\pi}{\lambda} (b+d) \sin \theta = \pm n\pi$$

$$(b+d) \sin \theta = \pm n\lambda$$

$$\text{When } n=0, \theta=0$$

The central max. of interference pattern lies along the direction of incident light. This is called principal maxima of zero order.

The central max. of diffraction pattern also lies along this direction. At this point all the waves arrive in same phase.

The ~~maximum~~^{minimum} intensity will be at,

$$\cos^2 \beta = 0 \quad \therefore \beta = \pm (2n+1) \frac{\pi}{2}$$

$$\frac{\pi (b+d) \sin \theta}{\lambda} = \pm (2n+1) \frac{\pi}{2}$$

$$(b+d) \sin \theta = \pm (2n+1) \frac{\lambda}{2}$$

It can be shown that for small values of θ , the maxima and minima are equally spaced. Thus $\cos^2 \beta$ term gives interference fringes analogous to those obtained in Young's double slit experiment.

If b is kept fixed, but $(b+d)$ is varied. The positions of maxima and minima due to diffraction remain unchanged while those of interference undergo a change.

